

Bifurcation Theory in One Dimension

Clean Notes, Phase-Line Interpretation, and Numerical Methods

Prepared for Students

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Overview

These notes introduce one-dimensional bifurcation theory through the geometry of

$$\dot{x} = f(x; r),$$

where x is the state variable and r is a control parameter. The emphasis is on how equilibria appear, disappear, exchange stability, and organize long-time dynamics. We first discuss the four standard local one-dimensional bifurcations: saddle-node, transcritical, supercritical pitchfork, and subcritical pitchfork. We then study a quintic example and the Spruce Budworm model in detail.

1 One-dimensional autonomous systems

Consider

$$\dot{x} = f(x; r), \quad x \in \mathbb{R}, \quad (1)$$

where r is a parameter.

An equilibrium x^* satisfies

$$f(x^*; r) = 0. \quad (2)$$

Thus, equilibria are precisely the intersections of the graph of $\dot{x} = f(x; r)$ with the horizontal axis.

1.1 The phase-line interpretation

The graph of $\dot{x}$ versus x immediately gives the direction of motion.

- If $\dot{x} > 0$, then x increases: trajectories move to the right.
- If $\dot{x} < 0$, then x decreases: trajectories move to the left.
- If $\dot{x} = 0$, then x is an equilibrium.

Consequently, stability can often be identified without calculation of a derivative: arrows pointing toward an equilibrium indicate stability, while arrows pointing away indicate instability.

1.2 Linear stability

Let

$$x(t) = x^* + \eta(t), \quad |\eta| \ll 1.$$

Linearization gives

$$\dot{\eta} = f_x(x^*; r)\eta.$$

Therefore,

$$\boxed{f_x(x^*; r) < 0 \Rightarrow x^* \text{ stable}, \quad f_x(x^*; r) > 0 \Rightarrow x^* \text{ unstable.}} \quad (3)$$

At a nonhyperbolic equilibrium,

$$f_x(x^*; r) = 0,$$

linear stability alone is inconclusive. Such points are precisely where local bifurcations can occur.

2 The four canonical one-dimensional bifurcations

2.1 Saddle-node (fold) bifurcation

The normal form is

$$\dot{x} = r - x^2. \quad (4)$$

Equilibria satisfy

$$x^* = \pm\sqrt{r}, \quad r \geq 0.$$

The positive branch is stable and the negative branch is unstable. At $r = 0$, the two equilibria collide and disappear.

The defining conditions are

$$f(x^*; r^*) = 0, \quad f_x(x^*; r^*) = 0, \quad (5)$$

with generic nondegeneracy conditions $f_{xx} \neq 0$ and $f_r \neq 0$.

2.2 Transcritical bifurcation

The normal form is

$$\dot{x} = rx - x^2 = x(r - x). \quad (6)$$

The equilibria are

$$x^* = 0, \quad x^* = r.$$

The two branches cross at $r = 0$ and exchange stability. This is the defining geometric feature of a transcritical bifurcation.

2.3 Supercritical pitchfork

The normal form is

$$\dot{x} = rx - x^3. \quad (7)$$

The equilibrium $x^* = 0$ is stable for $r < 0$ and unstable for $r > 0$. For $r > 0$, two stable equilibria

$$x^* = \pm\sqrt{r}$$

emerge. The pitchfork requires an appropriate symmetry, typically $x \mapsto -x$.

2.4 Subcritical pitchfork

The normal form is

$$\dot{x} = rx + x^3. \quad (8)$$

The zero equilibrium changes stability at $r = 0$, while the two nonzero branches

$$x^* = \pm\sqrt{-r}$$

exist for $r < 0$ and are unstable. In applications, subcritical pitchforks are often associated with abrupt transitions because the stable state can be lost before the parameter reaches the point at which a stable branch would emerge.

3 Worked quintic example

Consider

$$\dot{x} = rx + x^3 - x^5. \quad (9)$$

Equilibria satisfy

$$x(r + x^2 - x^4) = 0.$$

Thus $x = 0$, or with $y = x^2$,

$$y^2 - y - r = 0,$$

giving

$$y = \frac{1 \pm \sqrt{1 + 4r}}{2}, \quad x^* = \pm\sqrt{y}.$$

The derivative is

$$f_x = r + 3x^2 - 5x^4.$$

Using $r = y - y^2$ on the nonzero branches gives

$$f_x = 2y(2 - 3y).$$

Hence the nonzero branches are stable for $0 < y < 2/3$ and unstable for $y > 2/3$. This example illustrates how multiple folds can occur in a one-dimensional system.

4 Introduction: From Dynamical Stability to Bifurcation

Many physical, biological, ecological, and engineering systems evolve continuously in time. Although the underlying mechanisms may be complicated, their dynamics can often be represented by a differential equation of the form

$$\dot{x} = f(x; r), \quad (10)$$

where $x(t)$ is the state variable and r is a control parameter. The parameter r may represent a growth rate, interaction strength, external forcing, environmental condition, or any other quantity whose variation changes the behavior of the system.

The central question of dynamical-systems theory is not simply whether we can solve the differential equation explicitly. More importantly, we want to understand the *qualitative behavior* of its solutions:

Where does the system go?

Does the state approach a steady state? Does it move away from it? Can more than one long-term state coexist? What happens when a parameter is changed? Can a small change in the parameter cause a sudden qualitative transition?

These questions lead naturally to the concepts of *equilibrium*, *stability*, *bifurcation*, and *tipping*.

4.1 Equilibria as possible long-term states

An equilibrium x^* is a state that does not change with time:

$$\dot{x} = 0.$$

Therefore,

$$\boxed{f(x^*; r) = 0.} \quad (11)$$

If the system is placed exactly at x^* , it remains there forever. Equilibria are therefore natural candidates for long-term states of the system.

However, not every equilibrium is dynamically observable. The crucial question is what happens when the system is perturbed slightly away from it. Let

$$x(t) = x^* + \eta(t), \quad |\eta| \ll 1,$$

where η is a small perturbation. Linearizing gives

$$\dot{\eta} = f_x(x^*; r)\eta.$$

Hence

$$\eta(t) \approx \eta(0)e^{f_x(x^*; r)t}.$$

This immediately gives the fundamental one-dimensional stability criterion:

$$\boxed{f_x(x^*; r) < 0 \implies x^* \text{ is locally asymptotically stable,}} \quad (12)$$

whereas

$$\boxed{f_x(x^*; r) > 0 \implies x^* \text{ is unstable.}} \quad (13)$$

Thus stability is fundamentally a question of what happens to a small perturbation.

4.2 Stability from the geometry of $\dot{x}$

There is an especially intuitive way to understand stability in one dimension. For a fixed r , plot

$$\dot{x} = f(x; r)$$

against x .

If

$$\dot{x} > 0,$$

the state increases and moves toward larger x . If

$$\dot{x} < 0,$$

the state decreases and moves toward smaller x . Consequently, the zeros of the $\dot{x}$ curve are equilibria, while the sign of $\dot{x}$ determines the direction of motion.

For example,

$$\rightarrow \boxed{x^*} \leftarrow$$

indicates a stable equilibrium because trajectories on both sides move toward x^* . In contrast,

$$\leftarrow \boxed{x^*} \rightarrow$$

indicates an unstable equilibrium because trajectories move away from it.

Thus,

$$\boxed{+ \rightarrow - \text{ crossing: stable,} \quad - \rightarrow + \text{ crossing: unstable.}} \quad (14)$$

The phase-line criterion and the derivative criterion are two descriptions of the same local dynamics.

4.3 A Lyapunov viewpoint

Stability can also be formulated using a Lyapunov function. Instead of following the state directly, we construct a scalar quantity that measures the deviation from an equilibrium. For a one-dimensional equilibrium, a simple local choice is

$$V(x) = \frac{1}{2}(x - x^*)^2.$$

Clearly,

$$V(x) \geq 0, \quad V(x^*) = 0.$$

Its time derivative along trajectories is

$$\dot{V} = (x - x^*)\dot{x}.$$

Near the equilibrium,

$$\dot{x} \approx f_x(x^*; r)(x - x^*),$$

and hence

$$\dot{V} \approx f_x(x^*; r)(x - x^*)^2.$$

Therefore, if

$$f_x(x^*; r) < 0,$$

then

$$\boxed{\dot{V} < 0}$$

for sufficiently small perturbations away from x^* . The Lyapunov function decreases along trajectories, indicating that perturbations decay.

Thus, locally in one dimension,

$$\boxed{\text{negative slope} \iff \text{decaying perturbation} \iff \text{decreasing local Lyapunov function.}}$$

This provides a useful bridge between elementary phase-line analysis and the more general Lyapunov theory used for multidimensional dynamical systems. The quadratic function above is a convenient local construction; in general nonlinear systems, finding a suitable Lyapunov function can be considerably more difficult.

4.4 What changes when a parameter changes?

For a fixed value of r , a system may have one or several equilibria. Now imagine continuously varying r . The equilibria themselves can move:

$$x^* = x^*(r).$$

The collection of equilibrium states forms curves in the (r, x^*) plane. Such a curve is called an *equilibrium branch*.

An equilibrium branch is simply a continuous family of solutions of

$$f(x^*(r); r) = 0.$$

The important question is whether the number, stability, or qualitative organization of equilibria changes as r is varied.

4.5 What is a bifurcation?

A bifurcation occurs when a change in a control parameter produces a qualitative change in the dynamical structure of the system. This may involve

- creation of equilibria,
- destruction of equilibria,
- collision of equilibria,
- exchange of stability,
- appearance of multiple stable states, or
- a change in the number or type of attractors.

Thus bifurcation theory studies how the structure of a dynamical system changes as parameters are varied:

$$\boxed{\text{parameter variation} \longrightarrow \text{change in dynamical structure.}}$$

4.6 Stable and unstable equilibrium branches

An equilibrium branch is a continuous family of equilibria satisfying

$$f(x^*(r); r) = 0.$$

Each point on the branch has a stability determined by

$$f_x(x^*(r); r).$$

Thus,

$$\boxed{f_x < 0 \Rightarrow \text{stable}, \quad f_x > 0 \Rightarrow \text{unstable.}}$$

For the figures in these notes we deliberately use the following convention:

$$\boxed{\text{dashed curve} = \text{stable branch}}$$

and

$$\boxed{\text{solid curve} = \text{unstable branch.}}$$

This convention is used consistently in all x^* versus r diagrams. The dashed/solid distinction therefore carries dynamical information: it is not merely a visual distinction. At every value of r , a dashed point on a branch represents a stable equilibrium, whereas a solid point represents an unstable equilibrium.

4.7 How to read an x^* versus r bifurcation diagram

In an equilibrium bifurcation diagram,

$$\boxed{\text{horizontal axis} = r, \quad \text{vertical axis} = x^* .}$$

Each point on the diagram represents an equilibrium.

A continuous curve therefore represents a family of equilibria as r changes. The diagram answers the question:

$$\boxed{\text{How do the possible steady states change as the parameter changes?}}$$

The corresponding $\dot{x}$ versus x diagram answers a different question:

What does the system do for one particular value of r ?

The two plots should therefore always be interpreted together:

$$\dot{x} \text{ versus } x \Rightarrow \text{local dynamics and stability,} \quad (15)$$

while

$$x^* \text{ versus } r \Rightarrow \text{organization of equilibria and bifurcations.} \quad (16)$$

4.8 Saddle-node geometry

Consider

$$\dot{x} = r - x^2.$$

For $r > 0$,

$$x^* = \pm\sqrt{r}.$$

The positive equilibrium is stable and the negative equilibrium is unstable. As r decreases toward zero, the two branches approach one another and collide at

$$(r, x) = (0, 0).$$

For $r < 0$, no equilibria remain.

At the collision,

$$f(x^*; r^*) = 0, \quad f_x(x^*; r^*) = 0.$$

Thus the saddle-node is simultaneously an equilibrium and a point where the slope of the $\dot{x}$ versus x curve vanishes.

This explains why the x^* versus r diagram has a fold and why two equilibrium branches terminate at the same point.

4.9 Why bifurcations matter

Bifurcations are important because a system may undergo a dramatic qualitative change even when the governing parameter changes only slightly. For example, an ecological population can remain close to a low stable state while an environmental parameter changes slowly. At a critical parameter value, that stable state may disappear. The system is then forced to move toward a very different state.

Similar mathematical structures occur in ecological regime shifts, epidemic outbreaks, chemical reactions, neuronal transitions, synchronization, power-grid dynamics, and many other non-linear systems.

The central philosophy of these notes is therefore

understand the dynamics first; then understand how the dynamics change.

5 Geometric interpretation of the canonical bifurcations

The two most useful plots for teaching a one-dimensional bifurcation are:

1. the vector-field plot $\dot{x} = f(x; r)$ versus x for selected values of r ;
2. the equilibrium diagram x^* versus r .

The first plot explains the *dynamics and stability*; the second shows how equilibria are created, destroyed, or exchange stability as the parameter changes.

5.1 Saddle-node: $\dot{x} = r - x^2$

For $r < 0$ there are no equilibria. At $r = 0$, a double equilibrium occurs at $x^* = 0$. For $r > 0$, two equilibria appear:

$$x^* = -\sqrt{r}, \quad x^* = +\sqrt{r}.$$

The negative branch is unstable and the positive branch is stable.

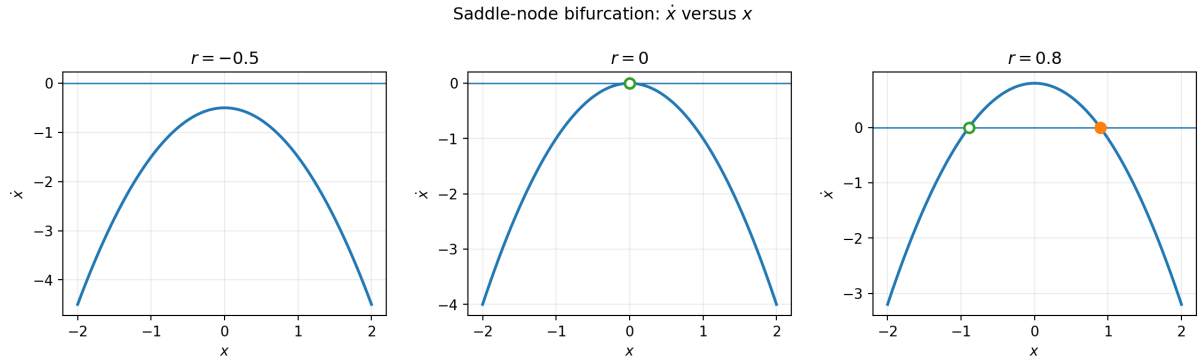


Figure 1: Saddle-node: $\dot{x}$ versus x for representative parameter values.

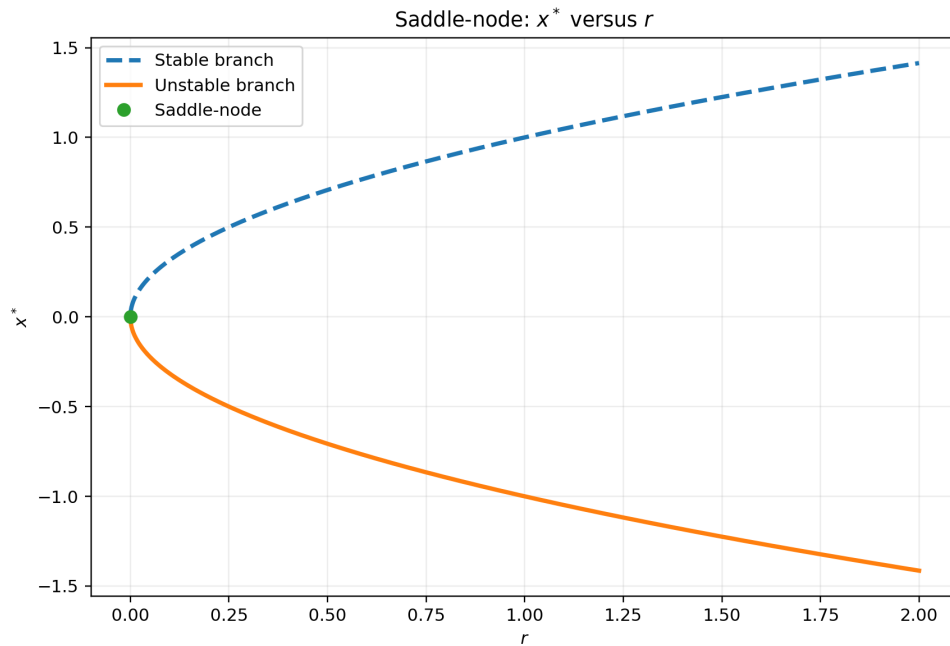


Figure 2: Saddle-node bifurcation diagram x^* versus r . Dashed curves indicate stable equilibria; solid curves indicate unstable equilibria.

5.2 Transcritical: $\dot{x} = rx - x^2$

The equilibria are

$$x^* = 0, \quad x^* = r.$$

The branches intersect at $(r, x) = (0, 0)$ and exchange stability.

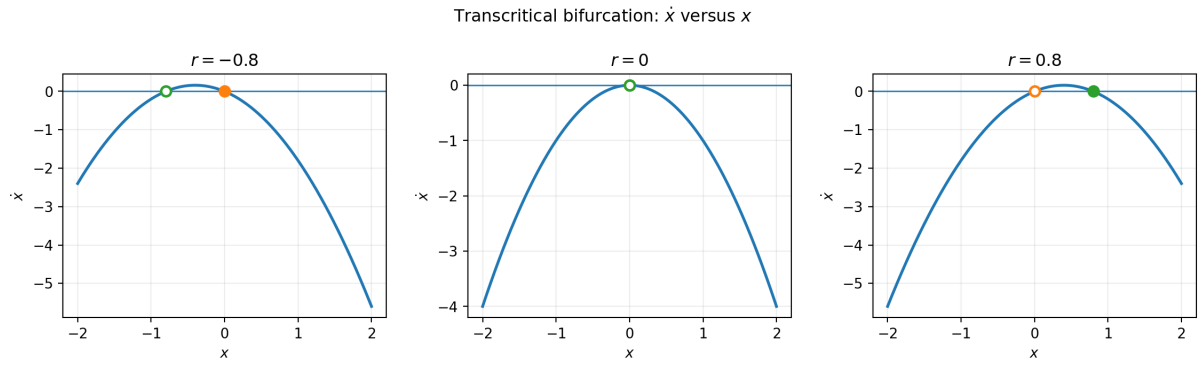


Figure 3: Transcritical bifurcation: $\dot{x}$ versus x .

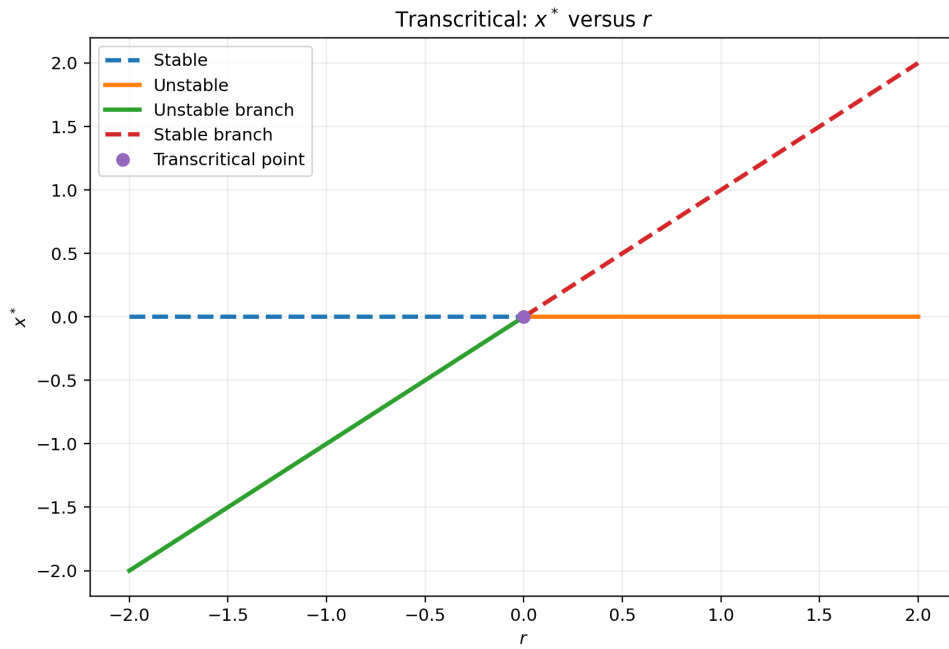


Figure 4: Transcritical bifurcation diagram x^* versus r .

5.3 Supercritical pitchfork: $\dot{x} = rx - x^3$

The equilibria are

$$x^* = 0,$$

and for $r > 0$,

$$x^* = \pm\sqrt{r}.$$

The zero branch is stable for $r < 0$ and unstable for $r > 0$; the two nonzero branches are stable for $r > 0$.

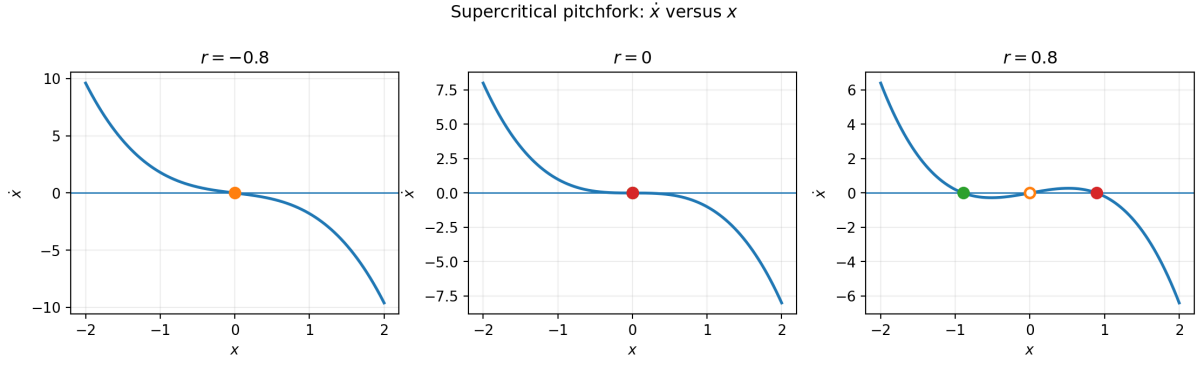


Figure 5: Supercritical pitchfork: $\dot{x}$ versus x .

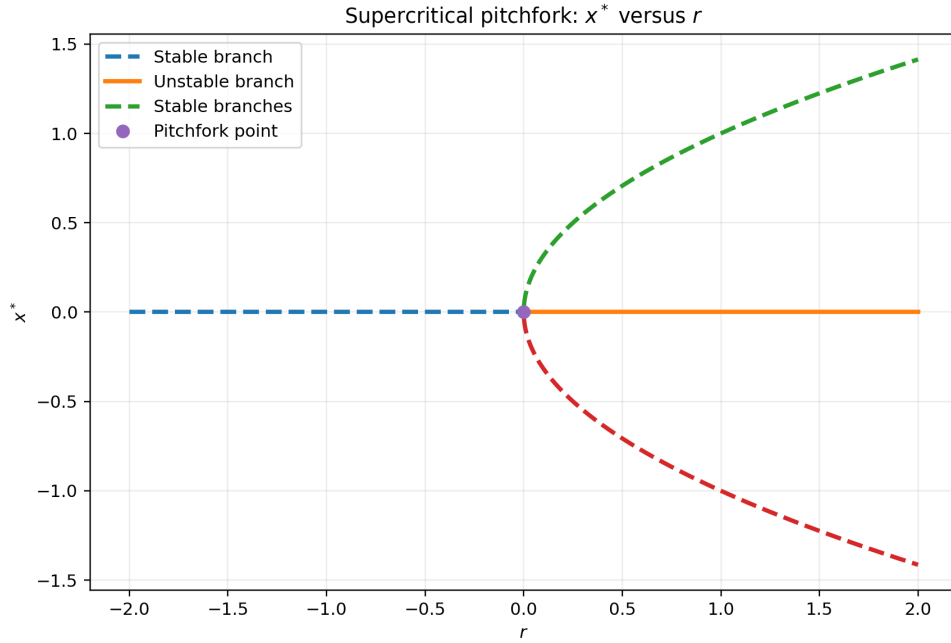


Figure 6: Supercritical pitchfork bifurcation diagram x^* versus r .

5.4 Subcritical pitchfork: $\dot{x} = rx + x^3$

The equilibria are

$$x^* = 0,$$

and for $r < 0$,

$$x^* = \pm\sqrt{-r}.$$

The two nonzero branches are unstable. The zero branch loses stability at $r = 0$. Unlike the supercritical case, the unstable nonzero branches exist on the side where the central equilibrium is still stable.

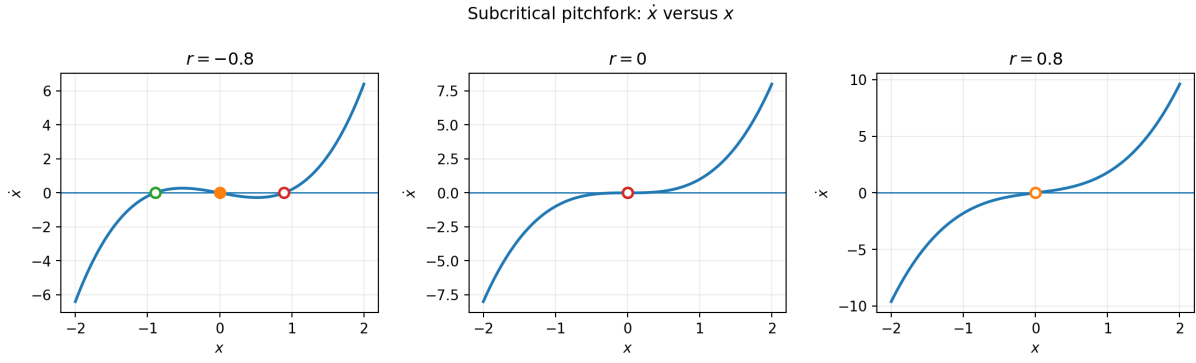


Figure 7: Subcritical pitchfork: $\dot{x}$ versus x .

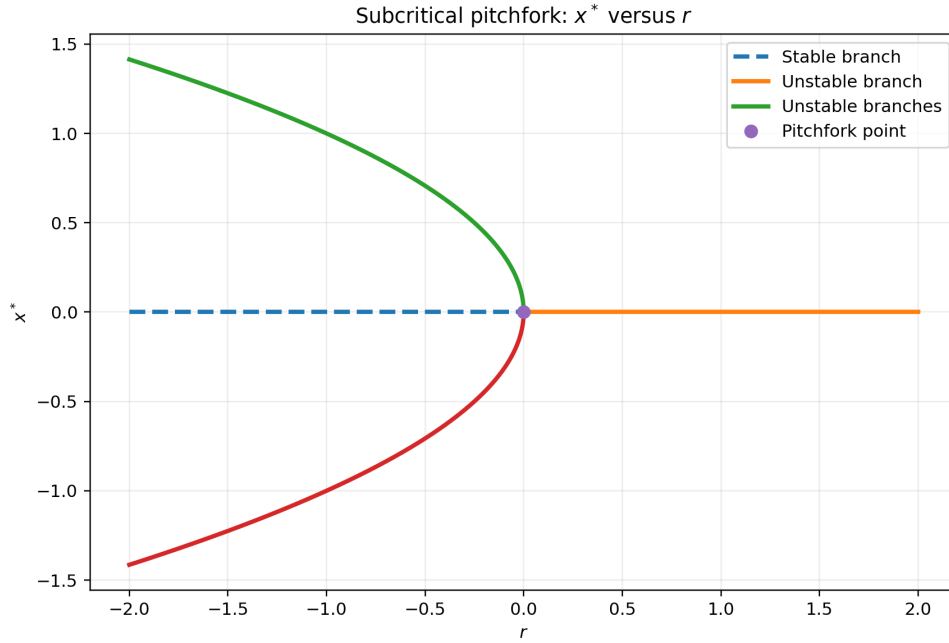


Figure 8: Subcritical pitchfork bifurcation diagram x^* versus r .

5.5 How to read the two figures together

For a fixed r , use the $\dot{x}$ versus x plot first: zeros give equilibria and the sign of $\dot{x}$ gives the direction of motion. Then use the x^* versus r diagram to see how those equilibria evolve as r is varied.

A useful rule is

$\dot{x}$: local dynamics and stability, $x^*(r)$: global organization of equilibria and bifurcations.

6 Spruce Budworm population model

We consider

$$\dot{x} = rx \left(1 - \frac{x}{K}\right) - \frac{x^2}{1+x^2}, \quad K = 10, \quad r > 0. \quad (17)$$

The first term represents logistic population growth, while the second is a saturating predation term.

6.1 Equilibria

Clearly,

$$x^* = 0$$

is always an equilibrium. Since $r > 0$,

$$f_x(0; r) = r > 0,$$

so the zero-population equilibrium is unstable.

For a nonzero equilibrium, divide by x :

$$r \left(1 - \frac{x}{K} \right) = \frac{x}{1 + x^2}.$$

Multiplication by $1 + x^2$ gives

$$r \left(1 - \frac{x}{K} \right) (1 + x^2) - x = 0.$$

For $K = 10$,

$$\boxed{-\frac{r}{10}x^3 + rx^2 - \left(1 + \frac{r}{10}\right)x + r = 0.} \quad (18)$$

Important: the nonzero equilibrium equation is cubic, not quadratic. This cubic is responsible for the possibility of three positive equilibria.

6.2 Stability

Differentiating Eq. (17),

$$\boxed{f_x(x; r) = r \left(1 - \frac{2x}{10} \right) - \frac{2x}{(1 + x^2)^2}.} \quad (19)$$

At every positive equilibrium:

$$f_x < 0 \Rightarrow \text{stable}, \quad f_x > 0 \Rightarrow \text{unstable}.$$

7 Why plot $\dot{x}$ versus x ?

For fixed r , plot

$$y = f(x; r).$$

The horizontal axis is x and the vertical axis is $\dot{x}$.

- Above the axis: $\dot{x} > 0$, so population increases.
- Below the axis: $\dot{x} < 0$, so population decreases.
- Axis crossings: equilibria.
- Crossing from $+$ to $-$: stable equilibrium.
- Crossing from $-$ to $+$: unstable equilibrium.
- Tangency to the axis: a candidate saddle-node point.

This plot is therefore a direct dynamical picture, not merely a mathematical graph.

8 Budworm results for the requested r values

Table 1: Equilibria and stability for $K = 10$.

r	Equilibrium x^*	$f_x(x^*; r)$ sign	Stability
0.25	0	+	Unstable
	0.259956	−	Stable
0.38	0	+	Unstable
	0.431231	−	Stable
0.50	0	+	Unstable
	0.683375	−	Stable
	2.000000	+	Unstable
	7.316625	−	Stable
0.56	0	+	Unstable
	7.727087	−	Stable

8.1 $r = 0.25$

There is one positive stable equilibrium:

$$x^* \approx 0.259956.$$

For small positive x , $\dot{x} > 0$; after the equilibrium, $\dot{x} < 0$. Therefore trajectories are attracted to the low-population state.

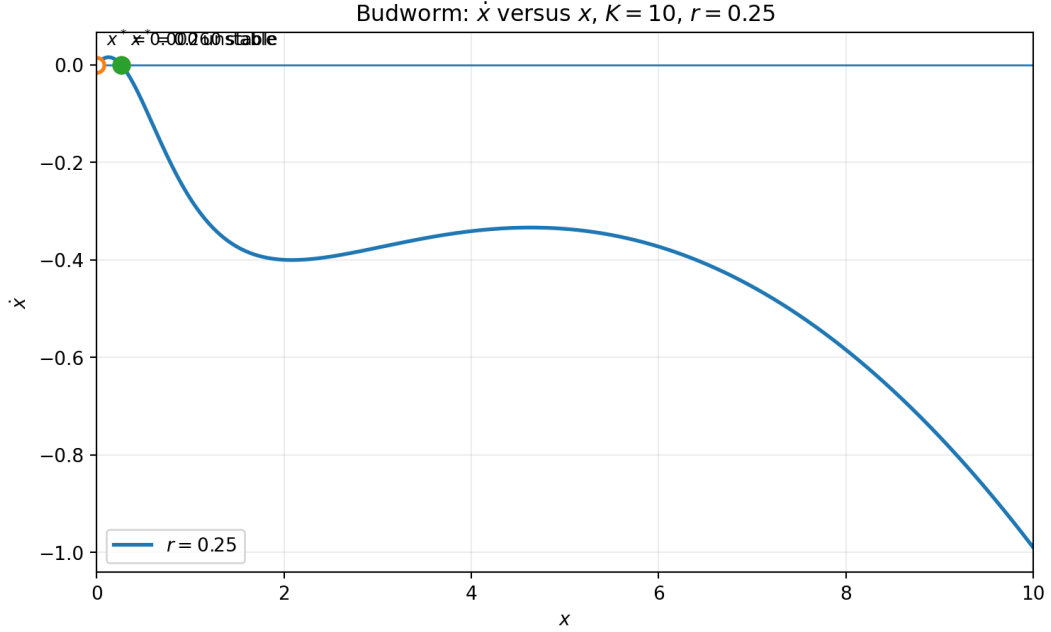


Figure 9: $\dot{x}$ versus x for $r = 0.25$.

8.2 $r = 0.38$: approaching the lower fold

The positive equilibrium is

$$x^* \approx 0.431231.$$

The lower saddle-node occurs near

$$r_{\text{SN},1} \approx 0.38397.$$

Thus $r = 0.38$ is close to the onset of the bistable region.

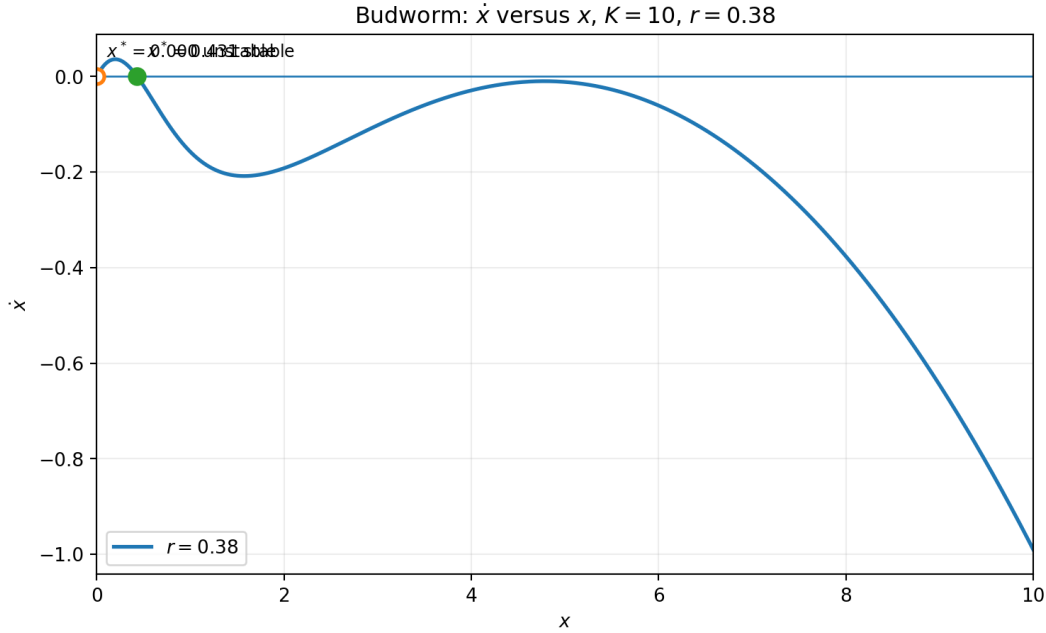


Figure 10: $\dot{x}$ versus x for $r = 0.38$.

8.3 $r = 0.50$: bistability

At $r = 0.50$ there are three positive equilibria:

$$x_1^* \approx 0.683375, \quad x_2^* = 2, \quad x_3^* \approx 7.316625.$$

Their stability sequence is

stable–unstable–stable.

The sign pattern is

$$\dot{x} > 0, \quad \dot{x} < 0, \quad \dot{x} > 0, \quad \dot{x} < 0$$

between consecutive equilibria. Therefore the arrows point toward the low and high equilibria and away from the middle equilibrium.

The middle equilibrium

$$x_2^* = 2$$

is an unstable threshold. In particular,

$$x(0) < 2 \Rightarrow x(t) \rightarrow 0.683375,$$

while

$$x(0) > 2 \Rightarrow x(t) \rightarrow 7.316625.$$

Thus the system has two attracting population states separated by an unstable threshold: this is bistability.

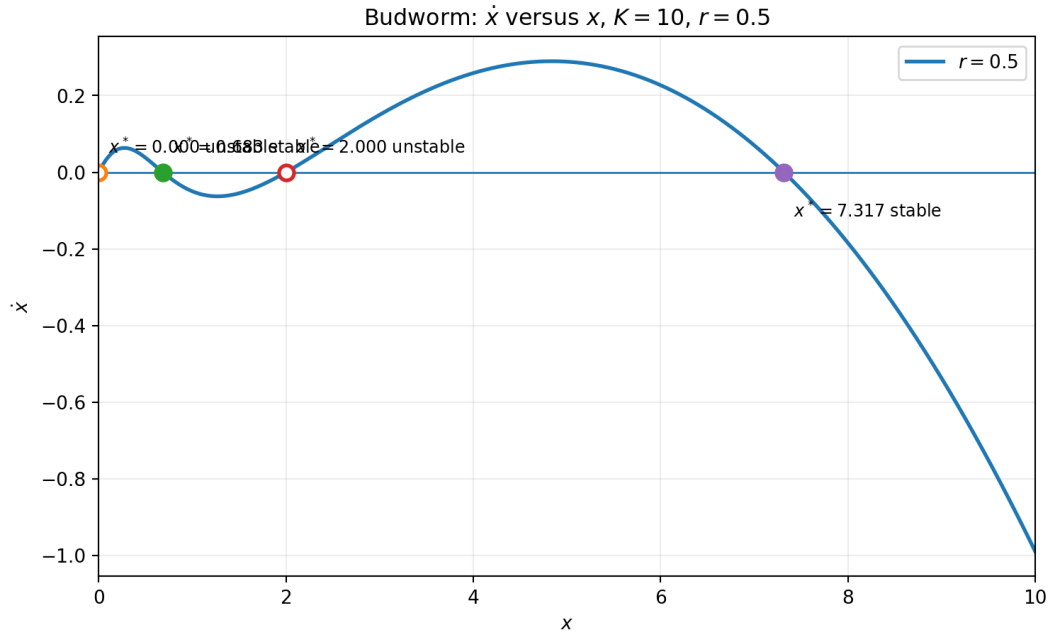


Figure 11: $\dot{x}$ versus x for $r = 0.50$, showing bistability.

8.4 $r = 0.56$: after the upper fold

The upper saddle-node is near

$$r_{\text{SN},2} \approx 0.55953.$$

Since $0.56 > r_{\text{SN},2}$, the low stable and middle unstable equilibria have disappeared. Only the high stable equilibrium remains:

$$x^* \approx 7.727087.$$

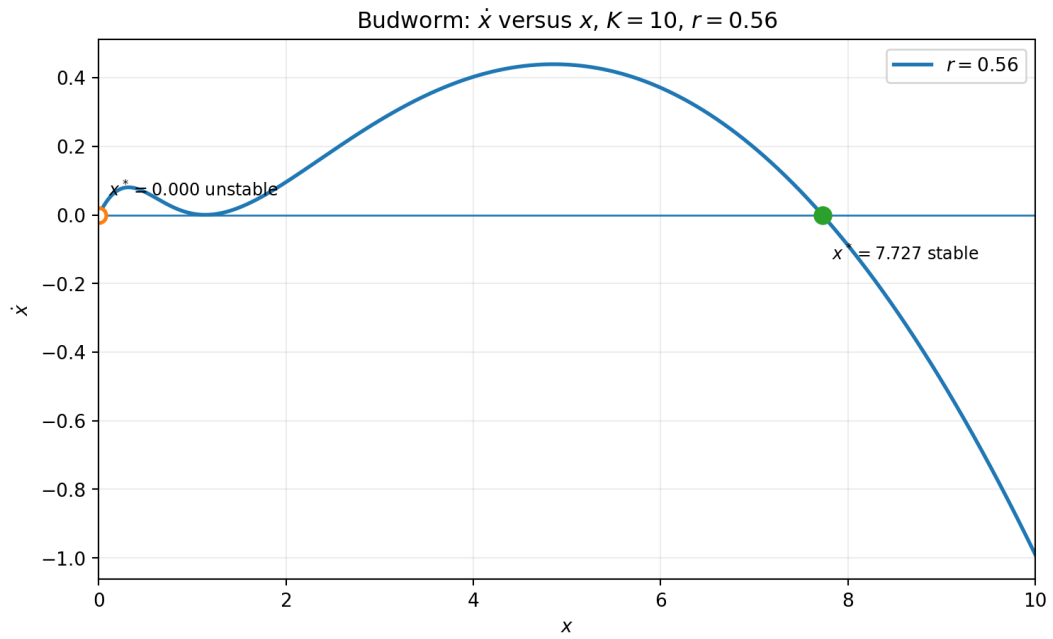


Figure 12: $\dot{x}$ versus x for $r = 0.56$.

9 Bistability and hysteresis

For approximately

$$0.38397 < r < 0.55953,$$

the model has three positive equilibria:

$$\boxed{\text{stable} - \text{unstable} - \text{stable}.}$$

The two stable states can be interpreted as a low-population refuge state and a high-population outbreak state. The unstable equilibrium separates their basins of attraction.

If r is slowly increased, the system can remain on the low stable branch until that branch terminates at the upper saddle-node. If r is then decreased, the system can remain on the high branch until the lower saddle-node is reached. This difference between upward and downward parameter sweeps is hysteresis.

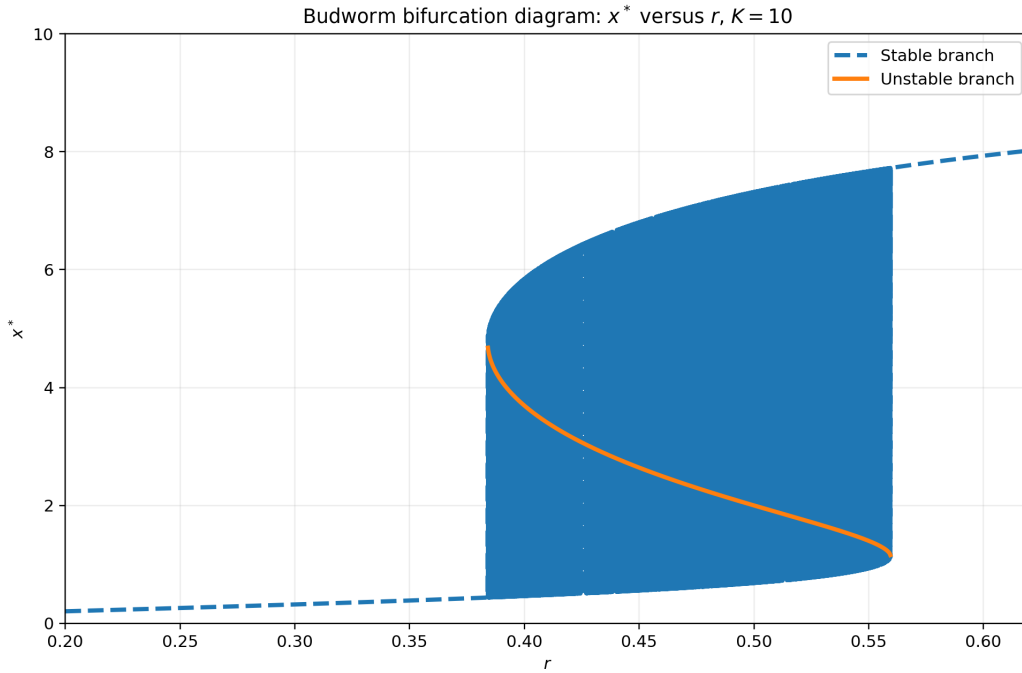


Figure 13: Numerically computed positive-equilibrium bifurcation diagram for $K = 10$. Stable equilibria are shown separately from unstable equilibria.

10 Fold detection

A fold is found by solving

$$\boxed{f(x; r) = 0, \quad f_x(x; r) = 0.} \quad (20)$$

For a generic saddle-node, one additionally requires

$$f_{xx}(x^*; r^*) \neq 0, \quad f_r(x^*; r^*) \neq 0.$$

For the Budworm model with $K = 10$, the two folds are approximately

$$\boxed{r_{\text{SN},1} \approx 0.38397, \quad r_{\text{SN},2} \approx 0.55953.}$$

11 Numerical workflow

For each requested value

$$r \in \{0.25, 0.38, 0.50, 0.56\},$$

students should:

1. Solve $f(x; r) = 0$ for all $x \geq 0$.
2. Check that no positive roots have been missed.
3. Calculate $f_x(x^*; r)$ at every equilibrium.
4. Mark equilibria stable or unstable.
5. Plot $\dot{x}$ versus x .
6. Draw arrows from the sign of $\dot{x}$.
7. Integrate the ODE from several initial conditions.
8. Repeat while slowly varying r to observe hysteresis.

11.1 Root finding

Newton's method is

$$x_{n+1} = x_n - \frac{f(x_n; r)}{f_x(x_n; r)}.$$

Because Newton's method can converge repeatedly to the same root, multiple initial guesses should be used. Bracketing methods such as bisection are useful for verifying all roots.

11.2 Time integration

Using RK4 or adaptive RK45,

$$\dot{x} = f(x; r),$$

start from several initial conditions. In the bistable regime, choose

$$x(0) = 1, \quad x(0) = 2, \quad x(0) = 3,$$

to demonstrate the role of the unstable threshold.

12 Summary

The one-dimensional equation

$$\dot{x} = f(x; r)$$

contains all the information needed to understand equilibrium dynamics.

1. Equilibria are zeros of f .
2. The sign of f determines the direction of motion.
3. A $+$ $\rightarrow$ $-$ crossing is stable.
4. A $-$ $\rightarrow$ $+$ crossing is unstable.
5. $f_x < 0$ gives stability and $f_x > 0$ gives instability.

6. $f = f_x = 0$ identifies a candidate fold.
7. Multiple stable equilibria imply multistability.
8. An unstable equilibrium between two stable equilibria acts as a threshold.
9. Saddle-node creation and destruction of equilibria naturally produce hysteresis.

For the Budworm model, the most instructive parameter is $r = 0.50$, because the $\dot{x}$ versus x plot directly displays the sequence

$$\boxed{\text{stable} - \text{unstable} - \text{stable}},$$

making bistability and the role of the unstable threshold visually transparent.